

Use vertically opposite angles

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle. Copy or complete the first step.

2. A triangle has angles 47 degrees and 68 degrees. Find the third angle.

3. A circle has radius 4.5 cm. What is its diameter?

4. Miro says: Miro estimates from the drawing even when the diagram is not to scale. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle. Copy or complete the first step.
Answer 68 degrees.
2. A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer 65 degrees.
3. A circle has radius 4.5 cm. What is its diameter?
Answer 9 cm.
4. Miro says: Miro estimates from the drawing even when the diagram is not to scale. Is this correct? Explain.
Answer Use labelled values and known geometric facts, then check the total angle relationship.

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Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

6. Check this result using an inverse, estimate or known fact: A circle has radius 4.5 cm. What is its diameter?

2. A triangle has angles 47 degrees and 68 degrees. Find the third angle.

7. Prove or explain your reasoning: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

3. A circle has radius 4.5 cm. What is its diameter?

8. Identify and correct this misconception: Miro estimates from the drawing even when the diagram is not to scale.

4. Solve this independently and show every stage: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A triangle has angles 47 degrees and 68 degrees. Find the third angle.

10. Write and solve a similar question to: A circle has radius 4.5 cm. What is its diameter?

Teacher answers and checking prompts

1. One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer 68 degrees.
2. A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer 65 degrees.
3. A circle has radius 4.5 cm. What is its diameter?
Answer 9 cm.
4. Solve this independently and show every stage: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer 68 degrees.
5. Use a second method or representation for: A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer Expected result: 65 degrees. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A circle has radius 4.5 cm. What is its diameter?
Answer Expected result: 9 cm. A valid check must be shown.
7. Prove or explain your reasoning: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
Answer 75 degrees. The explanation must show why the method works.
8. Identify and correct this misconception: Miro estimates from the drawing even when the diagram is not to scale.
Answer Use labelled values and known geometric facts, then check the total angle relationship.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A circle has radius 4.5 cm. What is its diameter?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

6. Create a new example that tests the same idea as: A circle has radius 4.5 cm. What is its diameter?

2. Prove the answer to this question, rather than only calculating it: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A triangle has angles 47 degrees and 68 degrees. Find the third angle.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 9 cm.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro estimates from the drawing even when the diagram is not to scale.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
Answer 75 degrees. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer Expected result: 68 degrees. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer Expected result: 65 degrees. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 9 cm.
Answer One valid reconstruction is: A circle has radius 4.5 cm. What is its diameter? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro estimates from the drawing even when the diagram is not to scale.
Answer Use labelled values and known geometric facts, then check the total angle relationship.
6. Create a new example that tests the same idea as: A circle has radius 4.5 cm. What is its diameter?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.